Introduction à la Programmation et l'Algorithmique

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Recursion

Recursion in a nutshell

- Usually, a function calls another function
- A function can itself
  - Recursive function
- Recursiveness is not appropriate to all problerms
  - Stack overflox
  - Prefer iterative solutions
- Recursion needs to define
  - A base case – when recursion must end
  - Recursive steps – recursive call with input

Recursion vs iteration

The analysis of the problem might lead to an iterative or recursive solution

$$n! = n \times (n-1) \times \cdots \times 2 \times 1 = \prod_{k=1}^{n} k$$

```
function factorial (n) {
  fact = 1
  for (i=1; i ≤ n; i++) {
    fact = fact ∗ i
  }
  return fact
}
```

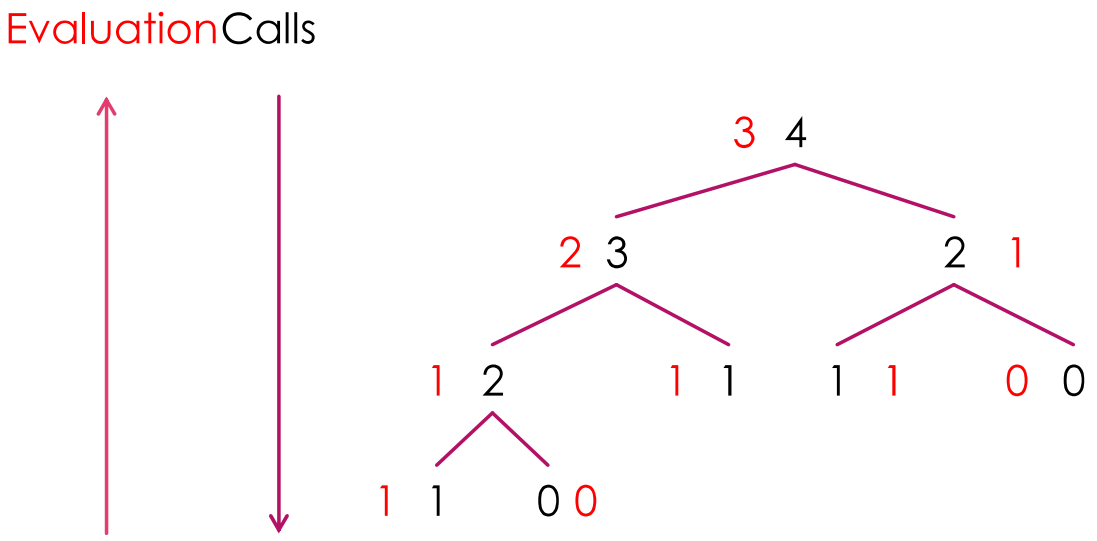
$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \times (n-1)! & \text{if } n > 0 \end{cases}$$

```
function factorial (n) {
  if (n == 0) {
    return 1
  } else {
    return n ∗ factorial(n−1)
  }
}
```

Avoid problem duplication

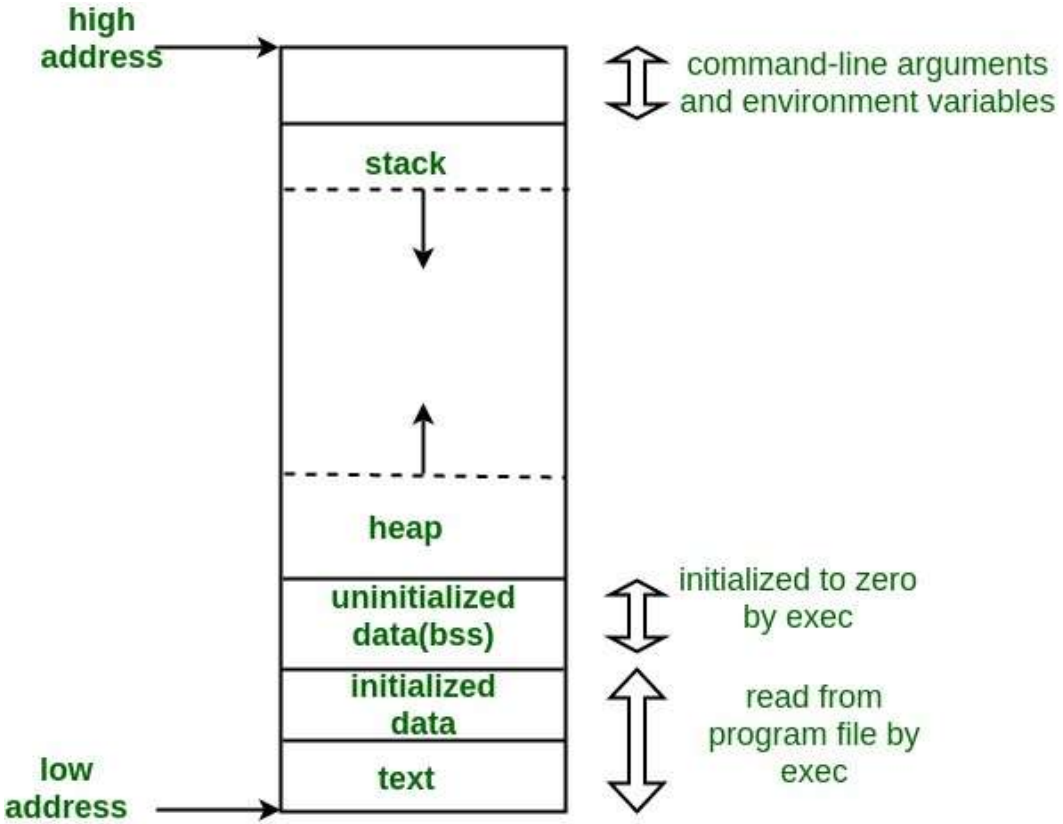
A bad Fibonacci recursive solution

```
function fibonacci (n) {
  if (n == 0) return 0
  if (n == 1) return 1
  return fibonacci(n-1) +
fibonacci(n-2)
}
```



A quick view of the virtual memory

And how the stack overflows



https://www.geeksforgeeks.org/memory-layout-of-c-program/